Stereoselective Reduction of 4-*tert*-Butylcyclohexanone with Sodium Borohydride

CM3191 • 22 October 2026 • Instructor: Dr Hoang Truong Giang
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1. Aim

4-*tert*-Butylcyclohexanone was reduced with sodium borohydride in 95% ethanol (Figure 1).[1,2] The *tert*-butyl group locks the ring in one chair conformation. Hydride can then add from the axial or the equatorial face. Axial attack gives the trans alcohol. Equatorial attack gives the cis alcohol. The reaction was followed by TLC. The product was isolated by aqueous work-up and extraction. The trans:cis ratio was measured from the ^{1}H NMR carbinol signals and used to assess the facial selectivity of $NaBH_4$.

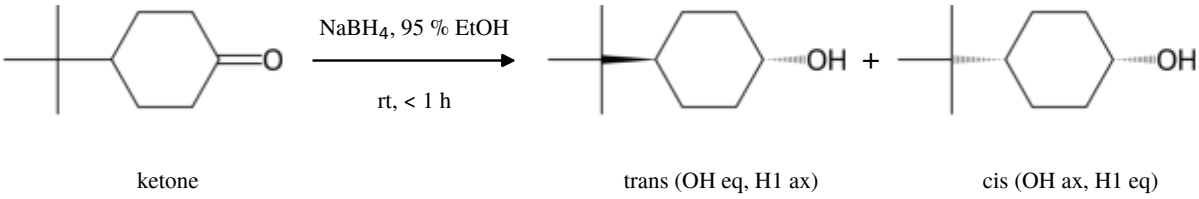


Figure 1. Reduction of 4-*tert*-butylcyclohexanone (MW 154.25) to 4-*tert*-butylcyclohexanol (MW 156.27). Axial attack gives the trans isomer (OH equatorial, H1 axial). Equatorial attack gives the cis isomer (OH axial, H1 equatorial).

2. Results

Table 1. Reagents.

Reagent	M / g mol^{-1}	Amount	n / mmol	Equiv.
4-*tert*-Butylcyclohexanone (limiting)	154.25	0.4929 g	3.195	1.00
$NaBH_4$	37.83	0.4040 g	10.68	3.34 (13.4 hydride)
Ethanol (95%)	46.07	5 mL	—	solvent

Table 2. Mass and yield, corrected for residual solvent by ^{1}H NMR (Appendix B).[3]

Flask + product	88.7095 g
Empty flask	88.3100 g
Product	0.3995 g
Product content	99.6 wt%
Corrected mass	0.3978 g
Theoretical yield	0.4994 g
Yield, uncorr. / corr.	80.0% / **79.7%**

Table 3. TLC (silica, hexane–EtOAc 4:1, front 4.0 cm, reaction lane near the end of the reaction, *p*-anisaldehyde stain). R_f uses the spot centre.

Lane	Spot / cm	R_f	Assignment
K (ketone)	1.5–1.7	0.40	ketone
R (reaction)	1.3–1.4	0.34	cis alcohol[a]
R (reaction)	1.0–1.1	0.26	trans alcohol[a]

[a] Tentative.

Table 4. Observations.

Stage	Observation
Ketone in 95% EtOH	Clear, colourless solution.
After $NaBH_4$ addition	White, cloudy suspension. Temperature not monitored. Stopped after just under 1 h.
Quench with water (3 mL) and 3 M HCl (1 mL)	Taken to the separatory funnel once a few drops of water caused no gas evolution.
Heating to boiling, cooling	Omitted on the TA's advice.
Extraction and washing	EtOAc (2 × 10 mL), water (2 × 10 mL), brine (10 mL), dried over Na_2SO_4.
Product after evaporation	White crystalline solid.

Table 5. ^{1}H NMR (500 MHz, $CDCl_3$, δ relative to $CHCl_3$ at 7.26)[4] and IR data. Integrals are relative to the sum of the two H1 signals = 1.00.

δ / ppm	Mult.	J / Hz	Int.	Assignment[a]
4.12	q	7.2	0.0135	EtOAc OCH_2
4.03	m (apparent quintet)	2.8	0.21	cis H1 (equatorial)
3.51	tt	10.9, 4.4	0.79	trans H1 (axial)
2.17	s	—	0.0078	acetone
2.06–1.96	m	—	1.65	trans H2/H6 eq
1.86–1.75	m	—	2.04	trans H3/H5 eq, cis ring H
1.57–1.44	br m	—	2.40	OH, H_2O, cis ring H
1.40–1.17	m	—	2.01	trans H2/H6 ax, cis ring H
1.08–0.93	m	—	2.73	trans H3/H5 ax and H4, cis H4
0.86, 0.84	2 × s	—	9.36	*t*Bu (cis, trans)
trans:cis 79:21 (H1 integrals), 80:20 (*t*Bu singlets, Lorentzian fit)				
IR (ATR, $\tilde{\nu}$/cm^{-1}) 3279 (br, O–H), 2940, 2858 (C–H), 1392, 1364 (*t*Bu), 1064 (C–O)				

[a] Ring assignments are tentative. Ring signals of the minor cis isomer overlap those of the trans isomer.

3. Discussion

3.1 Reaction monitoring

The reaction lane showed no spot at the ketone position (R_f 0.40). The ketone had therefore been consumed. Two more polar spots appeared at R_f 0.34 and 0.26. They are tentatively assigned to the cis and trans alcohols. An axial OH is more hindered and should bind silica less strongly.

The IR spectrum agrees with complete conversion. It shows no C=O band near 1715 cm^{-1}.[5] A broad O–H band appears at 3279 cm^{-1}. The strongest band, at 1064 cm^{-1}, fits a secondary alcohol C–O stretch.[5]

3.2 Product identity and purity

The ^{1}H NMR spectrum shows no signals near δ 2.2–2.4, where the ketone α-CH_2 protons would appear.[5] The *t*Bu integral of 9.36 fits one *t*Bu group per H1. Residual EtOAc and acetone account for 0.43 wt% (Appendix B).[3] The acetone probably came from rinsing glassware. Water overlaps the OH signal at δ 1.49 and was not quantified.

3.3 Stereoselectivity

The two H1 signals give the diastereomer ratio directly.[2]

$$\%\,\text{trans} = \frac{I_{3.51}}{I_{3.51} + I_{4.03}} \times 100\% = \frac{1.000}{1.000 + 0.258} \times 100\% = 79.49\%$$

The trans:cis ratio is therefore 79:21. A Lorentzian fit of the two *t*Bu singlets gives 80:20. The two methods agree.

The signal shapes confirm the assignment. The trans H1 is a triplet of triplets with J = 10.9 and 4.4 Hz. The large coupling comes from the two antiperiplanar axial neighbours. The cis H1 is a narrow multiplet with $J \approx 2.8$ Hz. All four of its neighbours are gauche. This pattern follows the Karplus relationship.[6] The equatorial H1 also lies 0.52 ppm downfield of the axial H1.

Hydride therefore added mainly from the axial face. Axial approach meets only the small 3,5-diaxial hydrogens. Equatorial approach suffers torsional strain with the axial C–H bonds at C2 and C6.[7] For a small nucleophile the torsional effect dominates. Larger nucleophiles shift attack towards the equatorial face. Axial attack falls from 88% for HC≡CLi to 0% for *t*BuMgBr.[2,7]

Hydride addition is irreversible, so the ratio reflects kinetic control. Taking 25 °C as the reaction temperature, it corresponds to

$$\Delta\Delta G^{\ddagger} = RT \ln \frac{[\text{trans}]}{[\text{cis}]} = 8.314\ \text{J mol}^{-1}\ \text{K}^{-1} \times 298\ \text{K} \times \ln \frac{1.000}{0.258} = 3.4\ \text{kJ mol}^{-1}$$

The trans isomer is also the more stable one, because its OH is equatorial. Its predominance here comes from the faster approach, not from equilibration.

3.4 Yield loss

The isolated yield was 80.0%. Correction for residual solvent lowered it to 79.7%. TLC and IR showed complete conversion. Most of the loss probably occurred during work-up. The boiling step was omitted. Some alcohol may therefore have stayed bound to boron as alkoxyborate in the aqueous phase. Some product also remained on the drying agent and glassware. The ethanol from the reaction stays in the aqueous phase and may carry some alcohol with it. Unquantified water would make the true yield slightly lower.

3.5 Improvements

A bulky hydride would reverse the selectivity. L-Selectride attacks from the equatorial face and gives mainly the cis alcohol.[8] $LiAlH_4$ is also a small hydride donor and should favour the trans alcohol. It needs dry ether solvents and a more careful quench. A lower temperature might raise the trans fraction slightly. This would need testing.

In principle 0.25 equiv of $NaBH_4$ supplies one hydride per ketone. The 3.34 equiv used supplied 13.4. Part of the reagent is consumed by the ethanol solvent with release of H_2.[2] A smaller excess would reduce H_2 evolution and borate waste. Its effect on reaction time would need testing. A co-spot lane on the TLC plate would confirm the absence of ketone. A longer NMR relaxation delay would make the integrals more rigorous.[3]

4. Conclusion

$NaBH_4$ reduced 4-*tert*-butylcyclohexanone completely. TLC, IR and ^{1}H NMR showed no remaining ketone. The alcohol was isolated as a white crystalline solid in 79.7% yield after solvent correction. The H1 integrals gave a trans:cis ratio of 79:21. The *t*Bu singlets gave an independent ratio of 80:20. Hydride therefore attacked the locked ketone mainly from the axial face. The H1 coupling patterns confirmed the axial and equatorial assignments.

References

[1] Hoang, T. G. *CM3191 Chemical Synthesis Experiments: Organic Lab Manual*; National University of Singapore, AY2026/27; Experiment 3, pp 14–18.

[2] Hoang, T. G. *CM3191 Org L3: Stereoselective Reduction of 4-tert-Butylcyclohexanone* (lecture slides); National University of Singapore, 2026.

[3] *Correcting Compound Mass and Yield for Residual Solvent Using ^{1}H NMR* (CM3191 handout); National University of Singapore, 2026.

[4] Fulmer, G. R.; Miller, A. J. M.; Sherden, N. H.; Gottlieb, H. E.; Nudelman, A.; Stoltz, B. M.; Bercaw, J. E.; Goldberg, K. I. NMR Chemical Shifts of Trace Impurities: Common Laboratory Solvents, Organics, and Gases in Deuterated Solvents Relevant to the Organometallic Chemist. *Organometallics* **2010**, *29*, 2176–2179.

[5] Pavia, D. L.; Lampman, G. M.; Kriz, G. S.; Vyvyan, J. R. *Introduction to Spectroscopy*, 5th ed.; Cengage Learning: Stamford, CT, 2015.

[6] Karplus, M. Contact Electron-Spin Coupling of Nuclear Magnetic Moments. *J. Chem. Phys.* **1959**, *30*, 11–15.

[7] Clayden, J.; Greeves, N.; Warren, S. *Organic Chemistry*, 2nd ed.; Oxford University Press: Oxford, 2012; p 828.

[8] Brown, H. C.; Krishnamurthy, S. Lithium Tri-*sec*-butylborohydride. A New Reagent for the Reduction of Cyclic and Bicyclic Ketones with Super Stereoselectivity. *J. Am. Chem. Soc.* **1972**, *94*, 7159–7161.

Data and Appendix

A Signed datasheet and IR spectrum

B Residual solvent and yield[3]

Integrals are relative to the sum of the two H1 signals (1H) = 1.00. The EtOAc quartet at δ 4.12 (2H) integrates to 0.0135. The acetone singlet at δ 2.17 (6H) integrates to 0.0078.

$$N_{\text{EtOAc}} = \frac{0.0135}{2} = 0.00675 \qquad N_{\text{acetone}} = \frac{0.0078}{6} = 0.00130 \qquad N_{\text{alcohol}} = 1.000$$

$$\text{wt\%}_{\text{alcohol}} = \frac{1.000 \times 156.27}{1.000 \times 156.27 + 0.00675 \times 88.11 + 0.00130 \times 58.08} \times 100\% = 99.573\%$$

$$m_{\text{pure}} = 0.3995\ \text{g} \times 0.99573 = 0.39779\ \text{g}$$

$$n_{\text{ketone}} = \frac{0.4929\ \text{g}}{154.25\ \text{g mol}^{-1}} = 3.19546\ \text{mmol} \qquad m_{\text{theor}} = 3.19546\ \text{mmol} \times 156.27\ \text{g mol}^{-1} = 0.49935\ \text{g}$$

$$\text{Yield} = \frac{0.39779\ \text{g}}{0.49935\ \text{g}} \times 100\% = 79.66\% \approx 79.7\%$$

$$n_{\text{NaBH}_4} = \frac{0.4040\ \text{g}}{37.83\ \text{g mol}^{-1}} = 10.679\ \text{mmol}$$

$$\text{Equiv.} = \frac{10.679}{3.19546} = 3.342 \qquad \text{Hydride equiv.} = 4 \times 3.342 = 13.37$$

C ^{1}H NMR spectrum (data set CM3191-Week 10-Thurs-5TBI-Oct22-2026-R09)

Name: ICLM Dylan

Lab Group Th. Fumehood B7 Date of the experiment: 2021/10/21, Th

Stereoselective Reduction of 4-tert-Butylcyclohexanone

Instruction: Take photos of the measurements. After completing the experiment, show the photos to your teaching assistant and ask them to sign your printed datasheet (hardcopy, not softcopy). Submit the signed datasheet along with your lab report

Starting materials

Mass of 4-*tert*-butylcyclohexanone added (g) 0.4929.

Mass of sodium borohydride added (g) 0.1040

Product

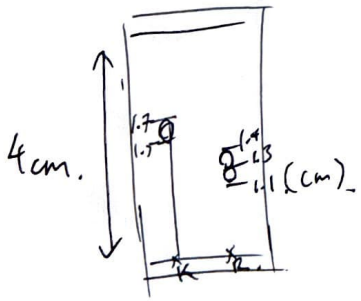
Mass of the empty rbf (g) 88.3100 g. ≈80%?.

Mass of product + rbf (g) 88.7095

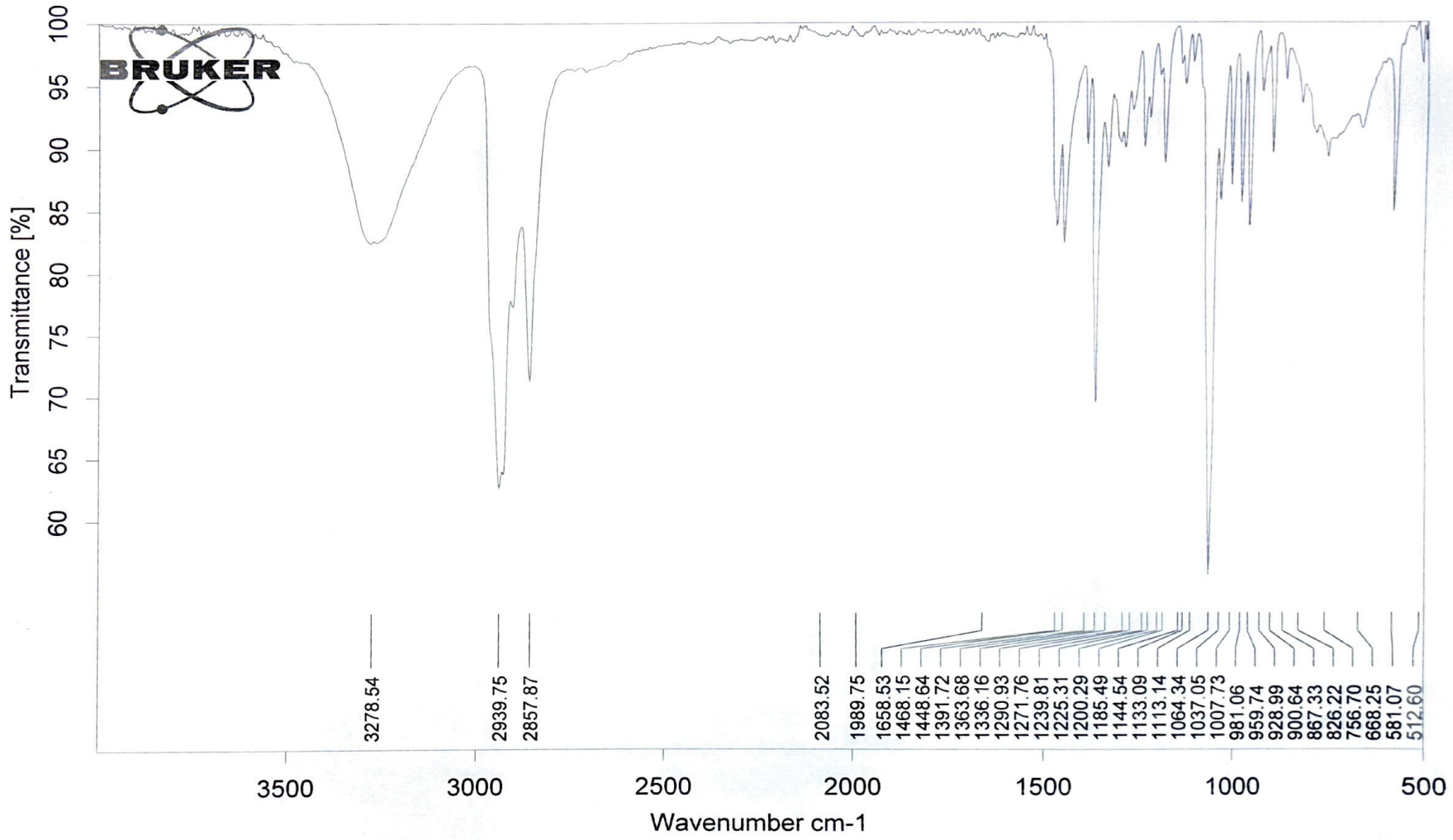
Mass of your final product (g) 0.3995

Observations

Draw your TLC plate here and label all distances (in cm)

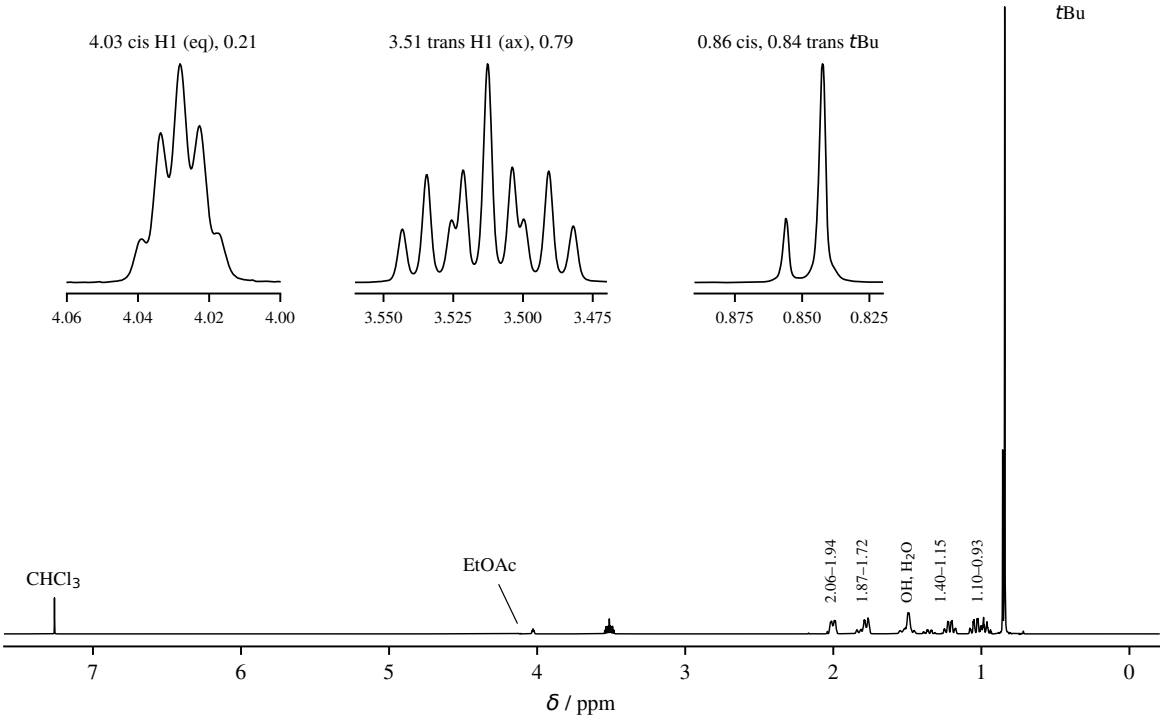


Sippakorn M.



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Sippakorn M.

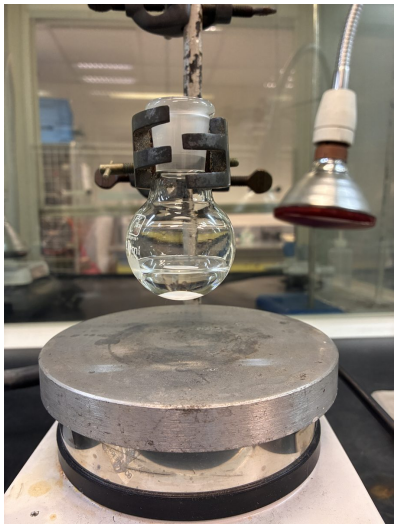


C ^{1}H NMR spectrum of the isolated product (500 MHz, $CDCl_3$). Insets show the two H1 signals and the two *t*Bu singlets.

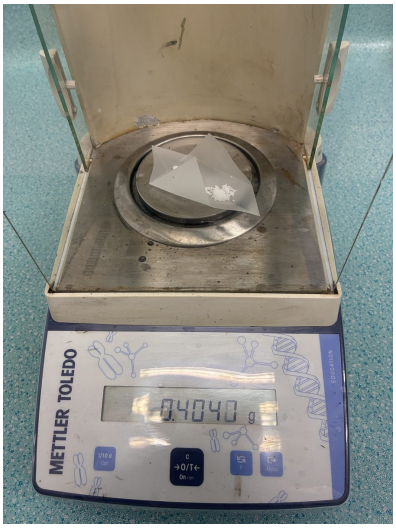
D Measurement photos and TLC plate



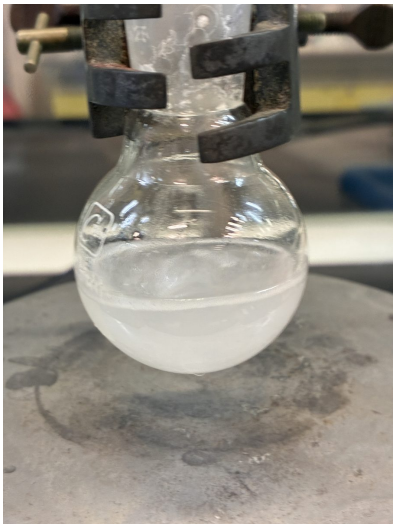
D1 Ketone, 0.4929 g



D2 Ketone in 95% EtOH



D3 $NaBH_4$, 0.4040 g



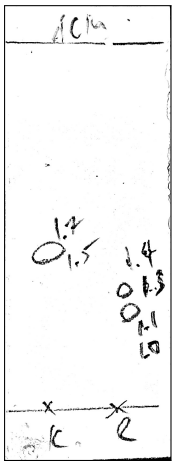
D4 After $NaBH_4$ addition



D5 Empty flask, 88.3100 g



D6 Flask + product, 88.7095 g



D7 TLC plate (hexane–EtOAc 4:1, *p*-anisaldehyde stain). Lane K is the starting ketone and lane R the reaction mixture. Distances are from the baseline in cm. The solvent front is at 4.0 cm. The photo was converted to greyscale and contrast-enhanced so the pencil marks show clearly.